Certified Clinical Engineer (CCE)

What is it, who runs it, how to earn it ?

Certified Clinical Engineer (CCE):

- An international / American certification granted to clinical engineers who successfully pass a standardized examination administered by professional boards (such as the Healthcare Technology Certification Commission in collaboration with AAMI and ACCE).
- It validates advanced knowledge and competence in the management, safety, and application of medical devices and healthcare technology.

Who Is Behind CCE?

- ACCE (American College of Clinical Engineering):
  - nonprofit (inc. 1990)
- HTCC (Healthcare Technology Certification Commission):
  - manages CCE program
- U.S. & Canadian Boards of Examiners:
  - set eligibility & exams
- *No membership required to pursue certification*

ACCE: Snapshot & Mission

- Represents professional interests of clinical engineers
- Sets standards of competence & promotes excellence
- Advances safe, effective tech in patient care
- Defines the profession's Body of Knowledge
- Official Website : https://accenet.org/

ACCE History

- 1989: Forum proposes new society (AAMI meeting, St. Louis)
- 1990 (Feb/May): Organizational meetings
- 1990 (June): ACCE incorporated (Washington State)

What Is a Clinical Engineer?

- “Supports and advances patient care via engineering + management of healthcare tech”

  (ACCE, 1992)Bridge between medicine, IT, and business

- Roles across hospitals, industry, academia, and consulting

CE vs. BMET (Clear Difference)

- **CE:** planning, evaluation, integration, management, compliance, projects
- **BMET:** inspection, preventive maintenance, repair

Why Certification? (The “Why”)

1. Formal recognition of competence & credibility
2. Standardized benchmark for employers & public
3. Drives patient safety and quality of care
4. Encourages lifelong professional growth
5. Better opportunities in larger healthcare hospitals and institutions

- *https://accenet.org/CECertification/Documents/Individuals_Certified_in_Clinical_Engineering_April2025.pdf*

Benefits to You

1. Stronger career mobility in hospitals, industry, government
2. Signaling to leadership & regulatory stakeholders
3. Access to study guides, review courses, and peer community
4. Part of a recognized professional pathway

Governance & Integrity

- Commission manages program; Boards design exams & renewals
- ACCE provides admin/financial support (not exam control)
- ACCE study resources are independent of exam writers

Current Program Status

- Modernization in progress
- New ACE/CCE applications: delayed until Q4 2025Updated written exam: expected Q1 2026
- Join notification list to receive 2025 handbooks/updates

Certification Process (3 Steps)

- Application review (eligibility, transcripts, 3 references)
- Written exam (computer-based; Prometric, global cities)
- Oral exam (professional judgment & depth)
- All communication/exams in American English
- Possible to be done online

Eligibility

- Option 1: Professional Engineer (PE) license + at least 3 years of clinical engineering experience.
- Option 2: Bachelor's degree in engineering (from an ABET-accredited institution) + 4 years of engineering practice + at least 3 years of clinical engineering experience.
- Option 3: Bachelor's degree in engineering technology (from an ABET-accredited institution) + 8 years of engineering practice + at least 3 years of clinical engineering experience.

Eligibility



- The board may accept degrees from U.S. programs accredited by agencies other than ABET, or even non-accredited programs, at its discretion.
- The ACCE defines clinical engineering practice as:
  - Engineering practice within the healthcare delivery system , or
  - Engineering work in support of healthcare delivery and patient care.

What the Exams Cover (Nature)

- Medical device & technology management
- Patient safety, risk, and quality systems
- Regulatory & standards context (e.g., FDA/CE/IEC principles)Project, budget & leadership topics
- Clinical systems integration & cybersecurity awareness

What the Exams Cover (Nature)

The contents of the written examination include 150 multiple-choice questions associated with the following topics:

I.	Technology Management	31.7%
II.	Service Delivery Management	19.9%
III.	Product Development, Testing, Evaluation, and Modification	4.3%
IV.	IT/Telecom	6.8%
V.	Education of Others	8.0%
VI.	Facilities Management	5.7%
VII.	Risk Management/Safety	10.2%
VIII.	General Management	13.4%

CCE Oral Examination (US candidates):

- Covers three clinical engineering scenarios.
- Each scenario has 5–7 questions.
- Time allowed: 20 minutes per scenario.
- Scoring:
  - 75% → based on accuracy and content of responses.
  - 25% → based on clarity, organization, and professional verbal presentation.

How to Prepare (Study Path)



- Read the CCE Candidate Handbook
- Use the ACCE CCE Study Guide (independent authors)Consider ACCE review course & webinars
- Build cases from your own CE work for the oral exam
- Plan a 6–12 month study window (typical)

Fees & Refunds (Latest Posted)

- 2024 Application fee: $475 USD
- Retest fees: $175 (written) / $150 (oral)Refund if ineligible: $325 USD
- No refunds for failed exams; fees not transferred to new dates(Updated 2025 links/fees: “coming soon”)

Membership vs Certification

- Membership NOT required for CCEACCE membership offers: webinars, discounts (AAMI/HIMSS/IFMBE), journals, volunteering, networking
- Global membership fees tiered by country income levels

Where CEs Work (HTM Landscape)

- Hospitals & health systems (technology planning → end-of-life)
- Asset & service management firms
- Device manufacturers & integrators
- Government & international orgs (e.g., FDA, WHO)
- Consulting (planning, expert witness, program audits)

Typical CE Responsibilities

- Strategic tech planning & procurement evaluation
- Implementation, integration, & cybersecurity safeguards
- Training clinicians & technicians
- Incident/recall management & regulatory compliance
- Utilization analysis & replacement planning

Ethics & Professionalism

- CCEs follow a professional Code of Ethics
- Patient well-being and confidentiality first
- Commitment to evidence, safety, and continuous improvement

ACCE Community & Volunteering / Join as Volunteer

- Committees (Body of Knowledge, Advocacy, Education, Membership, etc.)
- Mentoring & ambassadorships
- Journal and publisher access benefits
- Looks great on your CV and accelerates learning

So;

- CCE : validated competence in healthcare technology management
- Benefits you, your organization, and—most importantly—patients
- Next step: get on the 2025–2026 update list and begin structured prep

List of Certified Clinical Engineers



- Individuals Certified in Clinical Engineering

CCE Certificate Number	Full Name	City, State	Renewal Expiration Date
272	Abdullah Alaqeel, CCE-US	Riyadh, Saudi Arabia	6/30/2025
228	Hussain Ali, CCE-US	New York, NY	6/30/2027
388	Syed Ali, CCE-US	Doha, ,Qatar	6/30/2026
374	Michael Ambrosi, CCE-US	West Roxbury, MA	6/30/2025
321	Julia Angel , CCE-US	Chelmsford, MA	6/30/2027
196	Anthony Angelo, CCE-US	Frederick, MD	6/30/2026
296	Charles Armato, CCE-US	Little Rock, AR	6/30/2025
375	Zachary Arose, CCE-US	Dayton, OH	6/30/2025
285	Kamal Atwat, CCE-US	Marietta, OH	6/30/2027
232	Shashi Avadhani, CCE-US	Kendall Park, NJ	6/30/2027
322	Leslie Baggesen, CCE-US	Oakland, CA	6/30/2027
289	Mohammad Baharvandy, CCE-R US	Richmond Hill, ON	Retired
319	Parisa Bahrami, CCE-CA	Ottawa, ON	6/30/2026
108	Craig Bakuzonis, CCE-E US	Newberry, FL	Emeritus


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Thank you